

Healthcare Education: Program Completions

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AI Programming Foundations Project

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Overview

This project builds a Python workflow to examine awards in nursing and related healthcare programs. It uses the NCES Integrated Postsecondary Education Data System (IPEDS), specifically the 2023 final/revised Completions file and institutional directory (National Center for Education Statistics [NCES], 2023; [download the completions dataset](#)). The notebook loads, cleans, groups, and visualizes the records without training a machine-learning model.

Dataset Description

The prepared CSV contains 37,828 records and 16 columns from CIP series 51, Health Professions and Related Programs, covering awards granted from July 1, 2022 through June 30, 2023. Each record represents an institution, program, major number, and award level rather than an individual student. Key fields include program code and name, award level, institution state, award counts, and reporting flags. Program and award names were linked from the official dictionary, while institution names and states were joined by UNITID from the directory. The extract covers 3,925 institutions before the analytical filters (NCES, 2023).

Workflow Description

Pandas loads the local CSV and displays the first rows, missing-value counts, and record-key checks. The `normalize_labels` function trims text labels, and `select_comparable_records` selects first-major awards at detailed levels in the 50 states and Washington, DC. The `summarize_awards` function produces grouped totals and shares for programs, award levels, and institution states. Three Matplotlib bar charts and their accompanying interpretations communicate the results. NumPy checks that total awards match the sum of the reported men and women counts. The final summary brings together the main findings, assumptions, limitations, and questions the data cannot answer.

Key Decisions and Assumptions

The notebook combines code, documentation, results, and interpretation in one runnable document. This applies Danchev's (2022) approach to transparent, reproducible analysis. The CSV is kept beside the notebook so ordinary execution does not depend on a live download. Original counts and reporting flags remain unchanged. Wilson et al. (2017) recommend preserving source data and recording transformations in code; the cleaning functions return copies instead of overwriting the input dataframe.

Whitespace normalization changes 1,091 labels. There are no missing values or duplicate record keys in the selected source fields, so no imputation or duplicate deletion is applied. The analytical selection removes 453 second-major records and 564 additional first-major records outside the geographical scope, leaving 36,811 rows from 3,838 institutions. The file already contains detailed award levels; the function explicitly allows those levels to guard against counting subtotals alongside their components. All 8,338 zero-award records in the analytical population remain. Zero means zero reported awards, not missing data or proof that a program ceased operating.

Results and Interpretation

The filtered records total 951,577 first-major healthcare awards. Figure 1 shows the ten largest individual program categories. Registered Nursing/Registered Nurse leads with 250,576 awards, or 26.3% of the total. This is one program category; combining all nursing specialties would answer a different question.

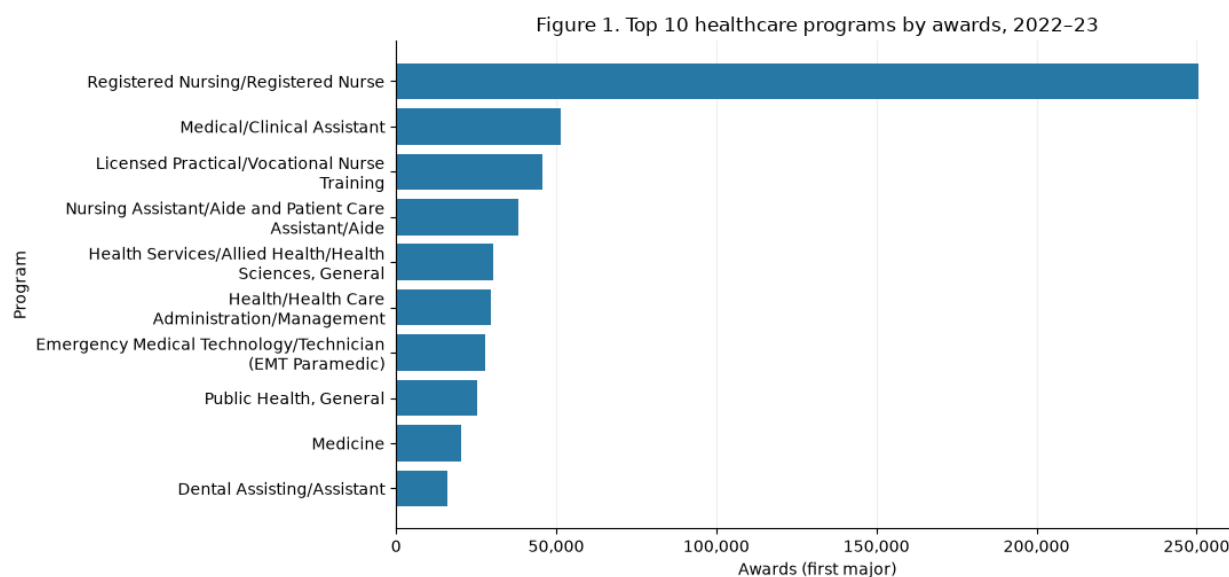
Figure 2 compares qualification levels. Bachelor's degrees are the largest category, with 251,787 awards, or 26.5%. Certificates and degrees represent different training pathways, so one award does not imply the same training duration or workforce contribution at every level.

Figure 3 shows the ten states with the highest counts. California leads with 97,017 awards. The locations belong to institutions, not students or future employers. These raw

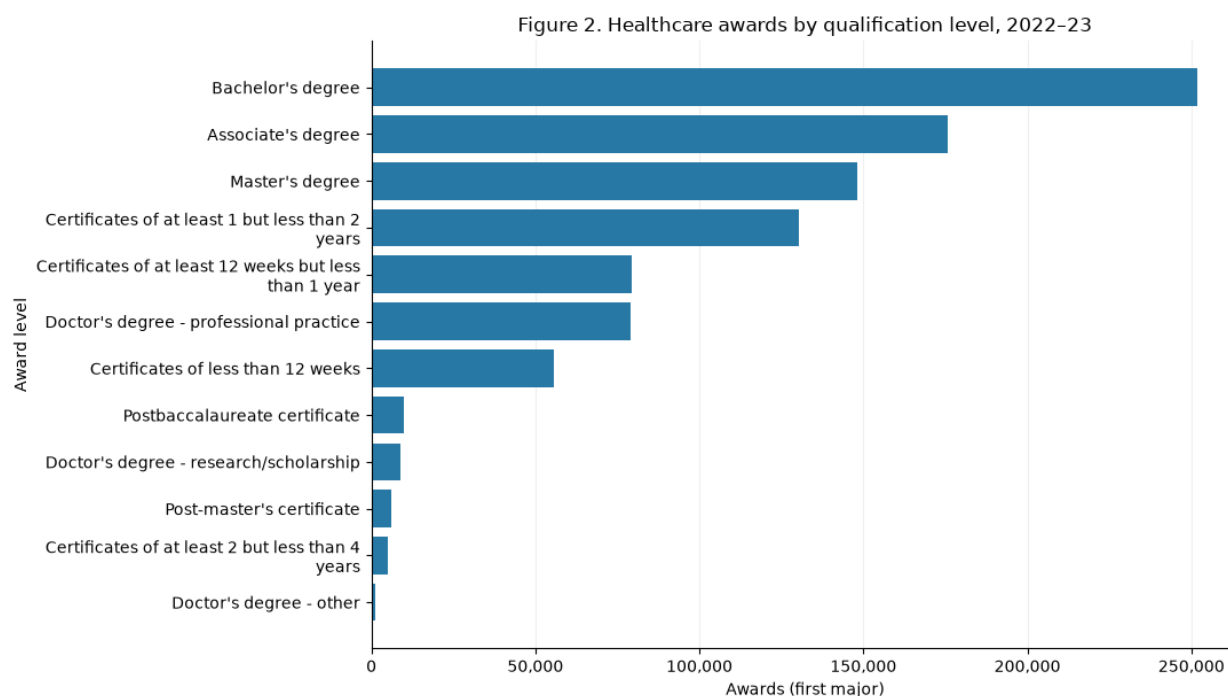
counts are not population-adjusted and should not be interpreted as rankings of education quality or access.

Figure 1

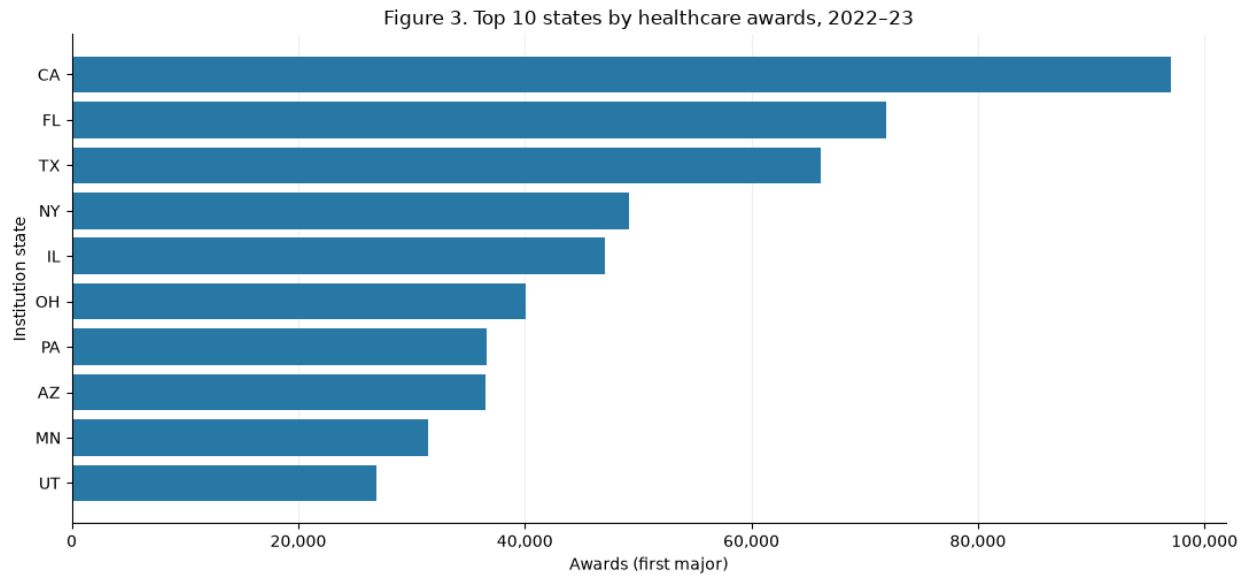
Ten Healthcare Programs With the Most Awards



Note. First-major awards in CIP series 51, 50 states and Washington, DC, July 2022-June 2023. Only the ten largest individual program categories are shown; percentages in the text use all 951,577 awards. Calculated from NCES (2023) data.

Figure 2*Healthcare Awards by Qualification Level*

Note. First-major awards in CIP series 51, 50 states and Washington, DC, July 2022-June 2023. All 12 detailed award levels in the analytical population are shown. Award levels represent different training pathways. Calculated from NCES (2023) data.

Figure 3*Ten States With the Most Healthcare Awards*

Note. First-major awards in CIP series 51, 50 states and Washington, DC, July 2022-June 2023. State refers to institution location. Counts are not population-adjusted and do not measure education quality. Calculated from NCES (2023) data.

Responsible Practice: Bias and Data Quality

The geographical filter excludes territories and freely associated states. That makes the scope consistent but means the findings cannot represent every location in the source. Restricting the analysis to CIP series 51 also excludes health-adjacent programs classified elsewhere. Dropping zero-award programs would raise average awards per record; treating missing counts as zero in a future release could lower totals. Those transformations are not performed here.

All total-award flags in this extract are R (reported), but reporting status does not eliminate possible reporting errors. Award counts do not identify unique graduates, licensure success, employment, or learning outcomes. A single reporting year cannot establish trends or causal effects. A next analysis could add population denominators, examine territories separately, and compare multiple years using consistent program definitions.

Why This Matters to My Work

I chose healthcare education because it connects directly to my research and the AI system I have built through [Learner Journey Labs](#). My work focuses on how students learn to work through clinical situations, understand what matters, and explain their decisions (Robinson, 2026). That makes the education behind those decisions important to me.

This project looks at a different part of that picture: where healthcare awards are being earned and which programs account for the largest share. These data give me a broader view of healthcare education, but they cannot tell me whether students are developing strong clinical reasoning or receiving the support they need. I want to keep that distinction clear. Completing a program and being prepared to apply what was learned are different questions.

Future Integration Reflections

Machine learning. A prediction task would need a defined outcome and an appropriate baseline. Predicting future award counts would require multiple years, a time-based holdout, and preprocessing fitted only on training data. The current one-year dataset supports description, not claims about future performance or student learning.

Neural networks. Categorical fields would need encoding, numeric inputs would need appropriate scaling, and the data would need conversion to tensors. Those transformations would be fitted on training data and then applied to validation and test data. A neural network would need to outperform simpler methods to justify its added complexity; this project does not establish that one is needed.

Agentic automation. An agent could check for a new IPEDS release, validate its columns, run the approved cleaning functions, and flag changes in totals. Human review would still be needed for changes in definitions, exclusions, and interpretation. Automation should not turn completion counts into unsupported claims about learning quality or readiness for clinical practice.

Reproducibility

The notebook ran top-to-bottom with 12 executed code cells, three embedded charts, and no saved cell errors. Grouped program, award-level, and state totals reconcile to 951,577 awards. The three chart images were visually reviewed. A macOS permission warning occurred during kernel shutdown after execution; it did not prevent the notebook from being saved.

The environment uses Python 3.12, with installed versions captured using pip freeze in requirements.txt. The README explains installation and execution. Version pinning follows Danchev (2022). The local main and development branches preserve separate stage commits, following Wilson et al. (2017). Project files and version history are available in the [GitHub repository](#).

References

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